

Performance and Reliability Tradeoff Analysis

Project: Indian Medical Insurance Corporation (IMIC)

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Executive Summary

This document provides a comprehensive and stylistically improved analysis of performance and reliability tradeoffs for the IMIC platform. It includes detailed architectural rationale, engineering considerations, operational guidelines, and implementation patterns to ensure a production-ready, scalable, and secure solution.

1. Introduction

The IMIC system consists of two primary digital applications: an Internet-facing platform for Policy Holders and Agents, and an Intranet-facing back-office for Insurance Company staff. These systems must support policy onboarding, claims processing, document management, and decision workflows while maintaining strict performance and reliability requirements.

2. Quality Attribute Deep Dive

- **Performance:** Ensures responsiveness in high-traffic areas like policy submission and claim intake. Includes response time targets, threading models, efficient DB queries, and caching strategies.
- **Reliability:** Guarantees correctness during underwriting and claim adjudication. Emphasizes transactional integrity, audit trails, and error-free processing.
- **Availability:** Targets continuous system operation (e.g., 99.9%+ uptime). Strategies include HA database clusters, backup policies, and resilient infrastructure.
- **Scalability:** Addresses anticipated load by supporting horizontal scaling for internet components and optimized vertical scaling for intranet layers.

- **Security:** Includes access control, encryption, audit logs, malware scanning for uploaded documents, and secure authentication with logout policies.
- **Consistency:** Critical for decision workflows. Ensures no partial approvals or mismatched claim states across modules.

3. Detailed Performance/Reliability Tradeoff Analysis

3.1 Policy Application Submission (Performance Priority)

Agent-driven policy submission is a throughput-sensitive workflow. The goal is to reduce latency and optimize customer onboarding. Below are architectural choices made to maximize performance.

- In-memory caching of premium/rating tables to accelerate calculations.
- Asynchronous file ingestion pipeline using queues.
- Client-side form validation to reduce server roundtrips.
- Optimized DB indexes for frequent read/write operations.

3.2 Policy Approval Workflow (Reliability Priority)

Approval processing involves validation of medical documents, identity verification, and underwriting rules. Reliability is prioritized to avoid incorrect policy activation.

- Full ACID transactions for multi-member policy approvals.
- Strict read-after-write consistency.
- No caching to ensure accurate real-time data visibility.
- Comprehensive audit logging for compliance operations.

3.3 Claims Management Workflow (Balanced Priority)

Claims involve both speed (submission) and reliability (adjudication). The architecture uses a hybrid strategy to balance performance and correctness.

- Event-driven claim intake using durable queues.
- Background processors to handle document processing workloads.
- Synchronous adjudication with strict consistency constraints.
- Idempotent APIs to prevent claim duplication under retries.

4. Architectural Patterns & Decisions

- CQRS separation for high-read traffic in public-facing modules.
- Event Sourcing Lite for claims and document submissions.
- Transactional Outbox pattern for reliable event publishing.
- Bulkhead isolation between Internet and Intranet components.
- Circuit breaker patterns for inter-service dependency protection.

5. Operational Excellence & Monitoring

- Define SLIs/SLOs for API latency, DB throughput, and claim processing time.
- Instrument distributed tracing for end-to-end workflow visibility.
- Enable proactive alerting for queue depth, DB locks, and error bursts.
- Adopt blue-green or canary deployments to minimize downtime.
- Enable structured logging (JSON) with correlation IDs.

Section 3 — Detailed Performance / Reliability Tradeoff

□ Performance-biased □ Reliability-biased □ Balanced

